



# AI and digitalization in relationship management: Impact of adopting AI-embedded CRM system

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## ARTICLE INFO

### Keywords:

AI  
CRM  
B2B relationship  
Technology turbulence  
Leadership support

## ABSTRACT

The purpose of this study is to determine the impact of adopting a artificial intelligence-embedded customer relationship management (CRM) system for business-to-business relationship management. After reviewing the literature and considering the theory, a conceptual model was developed. The model was validated using the PLS-SEM technique with 312 responses from 14 firms in the B2B context. The study finds that an AI-embedded CRM system has a significant positive impact towards B2B relationship satisfaction and firm performance. Also, the study highlights that there is a negative impact of the moderator 'technology turbulence' on the relations of 'automated decision making' and 'operational efficiency' with 'B2B relationship satisfaction', whereas there is a positive impact of moderator 'leadership support' on 'B2B relationship satisfaction' and 'firm performance'. There is presently no study on the impact of AI-CRM in the B2B context. Also, the study contributes to the existing literature by incorporating the moderator impact of 'technology turbulence' and 'leadership support' in the context of AI-embedded CRM systems for B2B relationship management.

## 1. Introduction

Customer relationship management (CRM) is considered a strategic macro process. It aims to ensure building and sustaining a profit-maximization portfolio concerning customer relationship with the firms (Saura, Palos-Sanchez, & Blanco-González, 2019). Many studies have been published on CRM in the B2C context, but explicit research on CRM in the B2B relationship context is still at a rudimentary stage (Irsic, 2017). B2B CRM activities help to analyze the customers' needs by examining the multifarious customer data (Kim, Park, Dubinsky, & Cha, 2012). Some of the huge-sized databases for such B2B relationship management are Matrix Marketing, Ampliz Salesbuddy, Easy Leadz, Leadspace, and so on (Ampliz, 2020). Firms use these B2B databases for accurate and faster targeting opportunities and cross-channel integration, and for effective collaboration with vendors and partners. Analysis of such a huge volume of customer data is difficult for humans. Therefore, appropriately using technology like artificial intelligence (AI) is essential (Josiasen, Assaf, & Cvelbar, 2014; Wang & Feng, 2012). This invites the need to study AI-integrated CRM systems in the B2B context (Harrigan, Soutar, Choudhury, & Lowe, 2015; San-Martina, Jiménez, & Lopez, 2016). AI-integrated CRM systems are associated

with automated decision-making capabilities, which also impact the operational efficiency of the firms in the B2B context, thus influencing satisfaction in the B2B relationship (Alshare & Lane, 2011; Hagen et al., 2020; Huang & Rust, 2018). With the help of AI-CRM, many routine activities can be automated to effectively reduce time compared to those conducted manually. Some of the routine tasks are automated forecasting, auto-processing data inputs, and retrieving and auto-determining the call list and response using NLP (Natural Language Processing) technology. Information transfer is the main ingredient of B2B relationship management (Michaelidou, Siamagka, & Christodoulides, 2011). It has to do with offering and receiving information, thus assisting firms to select and purchase B2B products or services (Agnihotri, Dingus, Hu, & Krush, 2016). There are several benefits that could be derived by using AI-CRM in B2B context. A few such benefits are automated partner selection, lead generation, appropriate segmentation, prioritization without manual intervention, and virtual assistance for vendor selection. An AI-integrated CRM system can help a firm automate its decision-making without human interference (Chatterjee, 2019). This hybrid system (AI-CRM system) can assist firms involved in B2B relationships to perform automated routine tasks and improve customization, segmentation, and prioritization of the acquired

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<https://doi.org/10.1016/j.jbusres.2022.06.033>

Received 2 November 2021; Received in revised form 8 June 2022; Accepted 15 June 2022

Available online 21 June 2022

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customer data. AI-CRM systems could eventually impact firms' performance (Chatterjee, Ghosh, & Chaudhuri, 2020a; Gotteland, Shock, & Sarin, 2020). However, AI is an emerging technology. Its integration with CRM in the B2B context might pose problems for the employees of the organizations to quickly adapt to AI-CRM technology. This rapid acquisition of a new technology might cause technological turbulence (Song, Dröge, Hanvanich, & Calantone, 2005; Agnihotri, Trainor, Itani, & Rodriguez, 2017). Conversely, the emergence of AI and ML systems in the B2B context has provided favorable support to the top management of the firms (Chatterjee, Ghosh, Chaudhuri, & Nguyen, 2019b; Thakur, Angriawan, & Summey, 2016). Rapid and unpredictable change in technology raises some issues with users, which is conceptualized as technology turbulence (Song et al., 2005). To use the new system, firms are required to quickly upgrade the necessary arrangements. This concept gives rise to the sense of technology turbulence (Ullah, Iqbal, & Shams, 2020). In the context of the present study, introducing AI-CRM systems in firms is found to be initially hindered by the users on various pretexts, which conforms to status quo bias (SQB) theory (Samuelson & Zeckhauser, 1988). The difference between just becoming automated and using AI in B2B CRM is that AI can recommend best available partners for a particular initiative, can provide virtual assistance for any vendor- or partner-related queries, can guide the team to plan and strategize a pragmatic roadmap towards improving lead generation, and can optimize opportunities, which could ultimately help the firm realize increased revenue. Nevertheless, there are no studies on how AI-CRM adoption for B2B relationship management impacts on automated decision making and how it influences the firms' operational efficiency, thus affecting firm performance. Nor is there any study on the moderating effects of technological turbulence and leadership support on firm performance in the context of an AI-integrated CRM system in B2B relationship management. To fill this gap, the aims of this study are as follows.

- [i]. To examine the impact of an AI-embedded CRM system on firm performance.
- [ii]. To determine how an AI-embedded CRM system impacts B2B relationship management.
- [iii]. To investigate the impact of 'technology turbulence' and 'leadership support' as moderators towards adoption of AI embedded CRM system for B2B relationship management.

## 2. Literature review

A huge volume of customer data can be effectively analyzed without requiring human interference by using an AI-integrated CRM system (Ferraris, Santoro, & Bresciani, 2017; Heide, Wathne, & Rokkan, 2007). In the B2B context, firms can use inputs from analyzing customer data to strengthen the CRM system (Herhausen, Miočević, Morgan, & Kleijnen, 2020; Kumar, Shankar, & Alijohani, 2019; Al-Omoush, Simón-Moya, Atwah Al-ma'aitah, & Sendra-García, 2021). B2B relationship marketing is interpreted as 'the marketing of goods and services to commercial enterprises, governments, and other non-profit organizations for use in goods and services they produce to sell to other industrial customers' (Chaston, 1999). A CRM system analyzes the volume of customer data, which a firm can use to effectively prioritize, customize, and efficiently segment those data (Alshaw, Missi, & Irani, 2011; Dalla Pozza, Goetz, & Sahut, 2018; Lacka, Chan, & Wang, 2020; Suoniemi et al., 2021). AI is useful for the CRM system to perform these activities in the B2B context (Chatterjee, Ghosh, Chaudhuri, & Chaudhuri, 2020b; Verma & Verma, 2013). When analyzing customer data through an AI-embedded CRM system, a firm's decision-making process is automated and more accurate, impacting its operational efficiency (Assimakopoulos, Papaioannou, Sarmaniotis, & Georgiadis, 2015; Greenberg, 2010; Maxwell, Jeffrey, & Lévesque, 2011). The use of AI-integrated CRM system in a firm impacts its performance provided there is strong leadership support (Inyang, Agnihotri, & Munoz, 2018; Thakur et al., 2016; Rana et al.,

2021). Several studies (Chatterjee, 2015, 2019; Lipiainen, 2015; Mustak, 2014) have observed that AI enhances the efficiency of a CRM system. Researchers (Ahearne et al., 2012; Nguyen & Waring, 2013) have also observed that applying new technology improves the business model through better communication activities, information control system and management design, thus improving CRM activities. Studies have revealed that applications of AI-integrated CRM systems impact firm performance (Coltman et al., 2011; Lee et al., 2014; Sreenivasulu, 2019; Piccolo et al., 2021). From the study of literature, it was observed that the CRM system effectively helps firms to improve their relationships with their customers (Kumar et al., 2019). Studies also have documented that firms with AI-embedded CRM are able to prioritize, customize, and segment the collected data of the customers, which eventually improves firm performance (Lipiainen, 2015; Lacka et al., 2020). However, from studying literature, it appears that no exhaustive study has investigated the impact of adopting AI-embedded CRM in a B2B context under the moderating influence of leadership support and technology turbulence. This has given rise to the need to study this aspect.

## 3. Theoretical background and development of hypotheses

### 3.1. Theoretical background

In the context of AI-CRM adoption for B2B relationship management, the firms need to emphasize how to utilize the available resources, especially various information of customers (Agnihotri et al., 2016; Chatterjee et al., 2020b; Davenport et al., 2020). In analyzing customer data with AI technology, the CRM system for B2B relationship management should improve firm performance. This requires firms to make best use of available resources. In this scenario, the management of the firm needs to involve its employees in improving the B2B relationships. Employees should learn how the available resources can help them make the B2B relationships more fruitful, effective, and dynamic (Fainshmidt et al., 2016). Here, it is deemed to be essential to nurture the resource-based view (RBV) theory (Kor & Mahoney, 2003) and dynamic capability view (DCV) (Teece et al., 1997). Both these theories posit that, to achieve better performance in the B2B relationship context with AI-CRM technology, firms need to develop and codevelop their appropriate sensing abilities. Therefore, they have to identify and assess the available and applicable technological opportunities to adequately mobilize the appropriate resources in dynamic B2B relationship management (Fainshmidt et al., 2016). Be it mentioned here that RBV is considered to present a managerial framework for determining the strategic resources a firm can exploit towards achieving sustainable competitive advantage. RBV focuses on firms' internal resources that are valuable, rare, inimitable, and not substitutable (VRIN). Proper utilization of these resources helps them to gain sustainable competitive advantage. In the context of B2B relationships, firms need to effectively deploy their VRIN resources to harness those resources between cross-functional units, alliance partners, and vendors. However, in the dynamic business environment, when customer demands are found to be changing rapidly, all firms involved in a B2B relationship should respond to those changing needs and articulate dynamic strategies amongst each other to gain joint success. This is the concept of DCV theory, which posits that firms need to possess the ability to integrate, build, and reconfigure internal and external resources to adapt to a rapidly changing business environment (Teece, 2012). Again, employees will pervasively and sometimes openly resist using any new system in a firm, as observed in terms of status quo bias (SQB) theory (Samuelson & Zeckhauser, 1988). The arrival of AI has also brought the emergence of the idea of technology turbulence. This is related with the concept that users of a new technology like AI or a new system commonly become inhibitors of it. They raise several excuses, such as they anticipate problems cropping up in the new technology, they doubt the user-friendliness of the new technology or the new system, and sometimes they are concerned about their readiness to use the

new technology (Dubey et al., 2019). In these circumstances, top management needs to convince the employees about the important benefits of applying the new technology, and management expectations to remove employees' initial hesitation (Schilke, 2014). The concepts of RBV theory and DCV help to explain how AI-CRM adoption for B2B relationship management can impact the quality of decision making and can enhance the firms' operational efficiency that, in turn, impacts its performance. Besides, SQB theory also helps to explain how technological turbulence can impede the AI-CRM technology adoption in the B2B relationship management context. However, leadership support can intervene to overcome many obstacles while adopting AI-CRM technology for B2B relationship management (Alshawhi et al., 2011; Chatterjee, 2019). Thus, RBV theory, DCV, and SQB theory could help to identify the antecedents that could eventually impact the firm performance.

### 3.2. Development of hypotheses and conceptual model

From studying literature and theories, we were able to identify the factors that would impact the firms' performance. In this section, we establish the relationship among these factors by formulating some hypotheses with consideration of two moderators 'technological turbulence' and 'leadership support'. These will help to develop a model conceptually.

#### 3.2.1. AI-CRM adoption for B2B relationship management (ACRM)

Firms that use AI-CRM in the B2B relationship management to analyze the customer data will be able to reduce the overall administrative costs for better marketing campaigns (Ascarza et al., 2018; Lipiainen, 2015). Firms that work together and use a CRM system in the B2B context will fetch better results by accurately taking appropriate decisions as to how to proceed with the analyzed customer data (Saura et al., 2019). But analyzing a huge volume of customer data manually is problematic, and here applications of AI come to rescue (Chatterjee et al., 2019). Thus, adopting AI-CRM for B2B relationship management could impact the decision-making system in an automated form that influences the operational quality and efficiency of the firms involved in B2B relationship (Mustak, 2014).

##### **Automated decision making (ADM)**

To improve the B2B relationship, the firms need to establish a close relationship with their customers and partners. The firms should collect various data on customers, partners, and clients to process through different CRM functionalities, to become useful and accurate information. The firms should then address their needs as and when required (Irsic, 2017). Managing, analyzing, and utilizing such huge volumes of data manually may not be possible, and the firm may fail to respond to the customers' need in an efficient and timely way. The manual decision-making process can also lead to employees' making erroneous decisions in managing their firms' B2B relationship with their customers, partners, and other clients. Thus, it is necessary to efficiently use such huge volume of customer data to arrive at an appropriate decision at the right time (Verma & Verma, 2013). Any kind of intelligent automation of a CRM system will help the employees better manage their B2B relationships. AI embedded with a CRM system can automate the decision-making process. This will also help to improve the satisfaction level of the employees involved in the B2B relationship (Alshare & Lane, 2011). It is also perceived that different stakeholders involved in B2B relationship management, such as managers and clients, will also experience improved levels of satisfaction. The above discussions help to formulate the following hypothesis.

*H1: AI-integrated CRM system enables automated decision making (ADM), which in turn helps to improve the B2B relationship satisfaction (BRS).*

##### **Operational efficiency (OPE)**

Firms need to innovate in multiple ways to improve their B2B relationships with customers, and an AI-integrated CRM system can complement those efforts. The firms collect customer data on their requirements, likings, dislikings (Li & Nguyen, 2016), and so on. Most of the time, the firms use different tools and processes to collect different kinds of customer data but managing a huge volume of data and using it at the right time in the B2B context is a challenge (Nguyen & Mutum, 2012). Firms collectively, synchronistical, and collaboratively synchronistical analyze customer data to devise the next course of actions (Lano, 2015). Most firms follow these usual operational procedures using a CRM tool to improve their B2B relationships. The more the firms can correctly analyze the data and can use the information at an appropriate time, the better they will be at maintaining B2B relationships. Adopting an AI-integrated CRM system may help in maintaining better relationships with firms' increased operational efficiency (Molinillo & Japutra, 2017). This is because the automated system can accurately manage and analyze huge volume customers without human involvement (De Luca, Herhausen, Troilo, & Rossi, 2020; Graca, Barry, & Doney, 2015; Rosenberg, 2018; Rust, 2020). As the firm's operational efficiency increases with the adoption of an AI-integrated CRM system, its B2B relationship will be positively impacted, which will increase the level of satisfaction of the people involved in B2B relationship management. This leads us to hypothesize as follows.

*H2: AI-integrated CRM system helps to improve the operational efficiency (OPE) of the firms, which in turn improves the B2B relationship satisfaction (BRS).*

The concept of AI-CRM adoption for B2B relationship management is considered an effective system to manage the firms involved in B2B relationships. This might also influence the firms' performance (Saura et al., 2019). Firm performance is conceptualized as how a firm can achieve its target in a more efficient way (Li, Ragu-Nathan, Ragu-Nathan, & Rao, 2006). Moreover, firm performance has a subjective measure, which corresponds to how well the firm manages its B2B relationships and how they influence its performance (Sin, Tse, & Yim, 2005). The firm performance can also be influenced by how well the firms' B2B relationships are measured, and the CRM tool plays an important part in it. The relationship can be improved if the firms' management can analyze customer information and needs quickly through proper segmentation and prioritization to arrive at an accurate decision quickly (Nunan, Sibai, Schivinski, & Christodoulides, 2018). This process takes less time and is more cost-effective when an automated system using AI technology is used to analyze the huge volume of customer data (Pandey & Mookerjee, 2018). Hence, a firm appropriately applying an AI-integrated CRM system towards managing the quality of B2B relationships can improve its performance. The AI-integrated CRM system can also automate a few processes such as understanding customers' requirements, their complaints, likings, and disliking. This also helps the firms to remove some of their manual tasks and to speed up the decision-making process, finally improving the firm performance (Inyang et al., 2018; Rodriguez & Honeycutt, 2011). The above discussions help to formulate the following hypothesis.

*H3: AI-CRM Adoption for B2B relationship management (ACRM) significantly and positively improves the firm performance (FP).*

#### 3.2.2. B2B relationship satisfaction (BRS) and firm performance (FP)

Firms track their efforts to improve their B2B relationships by using CRM software. AI technology complements their efforts (Chatterjee et al., 2019a). The AI-integrated CRM software combines a set of automated processes that are customer-oriented towards maximizing profit and revenue and improving B2B relationships to ensure better satisfaction of the people involved in B2B relationship management activities (Ahearne et al., 2012; Dans Perez, 2001; Davenport, Guha, Grewal, & Bressgott, 2020; Ma & Sun, 2020). Firms try to improve their B2B

relations, by becoming more sensitive and loyal using appropriate business strategy with the assistance of an AI-integrated CRM system (Firth & Swanson, 2001; Saura et al., 2019). Studies concerning B2B relationships using an automated AI-CRM system claim that relationship satisfaction is a key element towards assessing the health of customers' relationships with the firms (Chatterjee et al., 2019a). Improvement of B2B relationships also involves improving the firm's existing CRM system by integrating it with AI technology. Such integration also helps improve the firm performance (Chatterjee et al., 2019a; Rossomme, 2003). Firm performance is measured by the net outcomes, including consideration of both market and financial criteria, return on investments and assets, sales growth, market share, and overall profitability (Li et al., 2006; Sin et al., 2005). Level of satisfaction in the B2B relationship is also a measure of firm performance (Keramati, Mehrabi, & Mojir, 2010). The above discussions lead us to formulate the following hypothesis.

*H4: AI-integrated CRM system improves B2B relationship satisfaction (BRS) of the firms, which is positively related to firm performance (FP).*

### 3.2.3. Moderating effects of technology turbulence (TT)

If the relationship between two variables is not fixed, a third variable impacting on the relationship may strengthen the relationship, weaken the relationship, or even change the direction of the relationship. This third variable is called the moderating variable. Technology's rapid change and unpredictability is conceptualized as technology turbulence (Song et al., 2005). Because technology changes rapidly, the firms need to efficiently manage such changes. While incrementally upgrading an existing application, the firm might experience several issues. This is associated with the concept of technology turbulence (Ullah, Iqbal, & Shams, 2020). In the present study, technology turbulence in the B2B context is conceptualized as a situation concerning the emergence of AI technology and its integration with existing CRM applications (Santos, Borini, & Pereira, 2020). Business relations may undergo rapid changes when firms incrementally upgrade or introduce technology, especially if it changes the market orientation, impacting B2B relationships (Powell, Noble, Noble, & Han, 2018). It is possible to integrate AI technology with a firm's existing CRM system to improve and strengthen the B2B relationship. The employees who will use the AI-integrated CRM system in maintaining B2B relationships must be trained properly, otherwise, they could resist using it. This is also supported by SQB theory (Samuelson, & Zeckhauser, 1988). This is the concept of technology turbulence, and it is perceived to negatively affect B2B relationship satisfaction with its two predictors covering H1 and H2. These inputs lead us to formulate the following hypotheses.

*H5: Technology turbulence (TT) has moderating impacts in the relationship between automated decision making (ADM) and B2B relationship satisfaction (BRS) covering H1.*

*H6: Technology turbulence (TT) has moderating impacts in the relationship between operational efficiency (OPE) of the firms and B2B relationship satisfaction (BRS) covering H2.*

### 3.2.4. Moderating effects of leadership support (LS)

The use of AI technology with the existing CRM system to manage B2B relationships is considered a new idea. The AI-CRM application for B2B relationship management would impact on the firm's performance (Rossomme, 2003). But initially, its introduction could face resistance from the employees who are uncomfortable using the new system. It is in conformity with SQB theory. This is the concept of technology turbulence, as has already been discussed. In this situation, for firms to adopt the AI-integrated CRM system, employees need appropriate training (Chatterjee et al., 2020a; Lin, Yip, Ho, & Sambasivan, 2020; Rafiki, Hidayat, & Al Abdul Razzaq, 2019). Top management needs to actively support the adoption of the new system (Thakur et al., 2016). Leadership support and reinforcement will help align the employees to the firm's

strategy, influencing them to use the new system by removing any initial block to adopting the new system (Luo, Tong, Fang, & Qu, 2019; Sheth & Kellstadt, 2020; Vermeer, Araujo, Bernritter, Van Noort, 2019). Leadership can also support employees to overcome their initial unwillingness to use the new system. Thus, it is perceived that leadership support will have moderating effects towards the firm performance when using an AI-integrated CRM system which lead to hypothesize as follows.

*H7: Leadership support (LS) acts as a moderator to impact the relationship between the B2B relationship satisfaction (BRS) and the firm performance (FP) covering H4.*

All these hypotheses help to develop a conceptual model, as shown in Fig. 1.

## 4. Research methodology

To validate the proposed conceptual model, the partial least squares (PLS) structural equation modeling (SEM) technique was preferred. In terms of the observation of Hair, Howard, and Nitzle (2020), we note that the PLS-SEM technique is considered the most popular composite method for estimating structural equation modeling (SEM). This process comprises two parts. First is the research model in which we estimate the loadings of the items, and the average variance extracted, composite reliability, and Cronbach's  $\alpha$  of the constructs. The second part of this process is to estimate the predictive relevance of the model and to assess the model fit required for testing the hypotheses by SEM. This technique can estimate a complex model to provide robust results (Wetzels et al., 2009). The PLS-SEM approach was conducted with SmartPLS 3.2.3 software, and it considered non-parametric bootstrapping of 5000 replications to assess the path coefficients between the different linkages (Ringle et al., 2015; Akter et al., 2017). The PLS-SEM method can analyze data that are not normally distributed, which the CB-SEM technique cannot do (Rigdon et al., 2017; Willaby et al., 2015; Hair, Sarstedt, Ringle, & Gudergan, 2018). With PLS-SEM, we used a standard, 5-point Likert scale to quantify the responses from respondents in a survey that had a structured questionnaire.

### 4.1. Measurement instruments

It is necessary to prepare a set of questions (questionnaire) in the form of statements for the survey. The inputs from the literature review, the concepts of the constructs, and knowledge of the existing validated scale helped us to prepare the questionnaire. We prepared 32 questions (instruments) with a focus on ensuring that the respondents could understand the questions clearly, which would also help to improve the quality of responses. The questions were prepared to understand respondents' attitudes regarding AI-CRM adoption for B2B relationship management (Mellahi & Harris, 2016). Four experts who specialized in the domain of this study were asked for their help to fine-tune the questionnaire regarding its readability and comprehensiveness. Out of the four experts, two were from industry, each having more than 15 years of experience in B2B CRM activities. Of the other two experts, one has been a researcher on AI-CRM applications for more than 10 years. The other expert is an academican with more than 20 years of experience in CRM, ML, and Industry 4.0. The questions were prepared to be understandable so that there might not be a low response rate (Harzing et al., 2012). The response sheet provided five options for each question, and the respondents were asked to tick one of the options. A clear guideline was provided so that the prospective respondents would not find it difficult to accurately fill in the response sheet. The respondents were additionally informed that the aim of this study was purely academic, and that respondents' confidentiality and anonymity would be strictly preserved (Baurch & Holton, 2008; Chidlow, Ghauri, Yenyurt, & Cavusgil, 2015). Through all these editing processes, we attempted to limit the possible defects in the prepared 32 questions. Six measurement

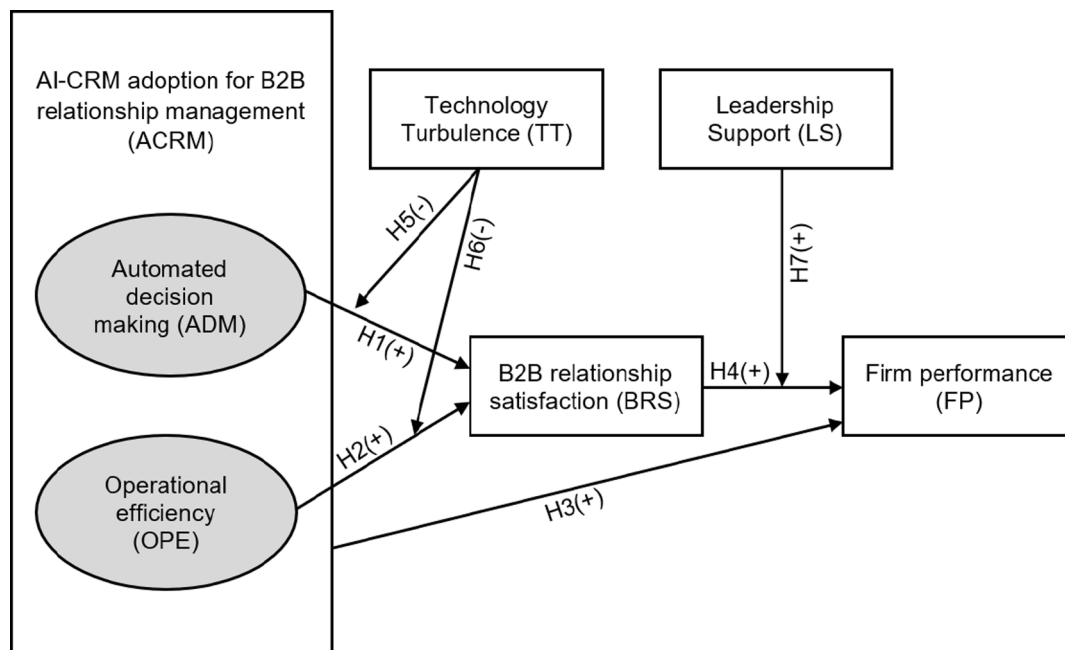


Fig. 1. The conceptual model.

items were framed to address automated decision-making (ADM), with inputs from different sources (Alshare & Lane 2011; Verma & Verma, 2013; Irsic, 2017; Chatterjee et al., 2019a; Davenport et al., 2020; Chatterjee et al., 2020ab). Five measurement items were prepared relating to operational efficiency (OPE), borrowing inputs from different sources (Nguyen & Mutum, 2012; Grace et al., 2015; Lano, 2015; Agnihotri et al., 2016; Li & Nguyen, 2016; Molinillo & Japutra, 2017). Eight measurement items were prepared relating to AI-CRM adoption for B2B relationship management (ACRM), with inputs from multiple sources (Alshawhi et al., 2011; Mustak, 2014; Lipiainen, 2015; Agnihotri et al., 2016; Ascarza et al., 2018; Chatterjee et al., 2019a; De Luca et al., 2020; Chatterjee et al., 2020a; Davenport et al., 2020). B2B relationship satisfaction (BRS) was measured with six items from different sources (Dans Perez, 2001; Firth & Swanson, 2001; Sin et al., 2005; Li et al., 2006; Keramati et al., 2010; Ahearne et al., 2012; Saura et al., 2019; Chatterjee et al., 2019a). Seven measurement items were articulated in connection with the construct of firm performance (FP), derived from inputs from different sources (Dans Perez, 2001; Firth & Swanson, 2001; Rossumme, 2003; Sin et al., 2005; Li et al., 2006; Keramati et al., 2010; Ahearne et al., 2012; Chatterjee et al., 2019a; Saura et al., 2019). In preparing the questionnaire, a step-by-step procedure was followed, as provided by Carpenter (2018). The procedure is illustrated in Fig. 2.

The details of the questionnaire, along with the corresponding sources, are provided in Appendix A1.

#### 4.2. Data collection strategy

To target potential respondents from the different Indian firms, a list of the firms was procured from the Bombay Stock Exchange (Mumbai, India). From the list, 34 firms were selected at random. The top executives of those firms were contacted by phone and by email. However, it was known that, out of these 34 firms, only 21 works in the B2B context with an AI-integrated CRM system. The top management of these 21 firms were also contacted by phone and email. They were requested to allow their managers to participate in the survey. The management of these firms were contacted more than once to convince them that the survey would be conducted for purely academic purposes and the identities of the respondents taking part in the survey would be kept strictly confidential. However, an initial reluctance was perceived from

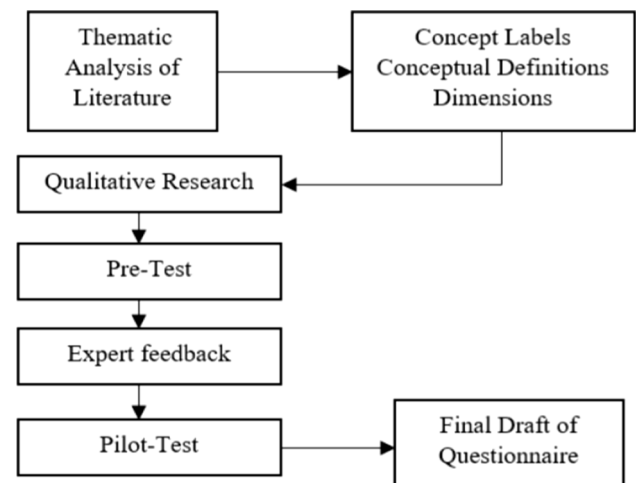


Fig. 2. Steps to prepare questionnaire (Source: Carpenter, 2018).

the side of the top management of these 21 firms. After the initial hiccup, eventually, it was possible to convince top executives of 14 firms to allow their managers of different hierarchy to respond to this survey. Only those managers who are actively involved in B2B relationship management were contacted. Out of 14 firms, 10 firms were large, whereas four were small and medium firms. Also, four firms were from the telecommunication sector, three from the IT sector, two from the retail sector, two from the pharmaceutical sector, and three from the automobile sector. The details are shown in Table 1.

In this way at the outset, 690 managers at these 14 firms were selected. They were provided with the questionnaire along with the instructions, and were asked to respond within two months, that is, January and February 2020. They were contacted over telephone and email more than once to return their responses within the stipulated time. Ultimately, by the first week of March 2020, only 326 responses from the managers of several ranks at those 14 firms were obtained, which is a response rate of 47.3%. To perform a non-response bias test, procedures recommended by Armstrong and Overton (1977) were

**Table 1**  
Demographic information.

| Particulars       | Category                   | Number | Percentage (%) |
|-------------------|----------------------------|--------|----------------|
| Working position  | Senior Manager             | 99     | 31.7           |
|                   | Midlevel Manager           | 147    | 47.1           |
|                   | Junior Manager             | 66     | 21.2           |
| Firm Type         | Service firm               | 9      | 64.3           |
|                   | Manufacturing firm         | 5      | 35.7           |
| Firm Size         | Large firm                 | 10     | 71.4           |
|                   | Small and medium size firm | 4      | 28.6           |
| Industry / Sector | Telecommunication          | 4      | 28.6           |
|                   | IT                         | 3      | 21.4           |
|                   | Retail                     | 2      | 14.3           |
|                   | Pharmaceutical             | 2      | 14.3           |
|                   | Automobile                 | 3      | 21.4           |

followed. A chi-square test and an independent *t*-test were conducted on the responses from the first 100 and the last 100 respondents. The results did not highlight any mentionable deviation, thus concluding that there was no non-response bias.

After verification of these 326 responses, it was found that 14 responses were defective because either the response sheets concerned were found blank or in some cases, the respondents put more than one tick mark for a single statement. These 14 responses were not considered. Analysis by PLS-SEM was done on 312 usable responses to 32 questions, which is considered acceptable (Deb & Lomo-David, 2014). The demographic cluster information of these 312 respondents is given in Table 1.

## 5. Data analysis and results

### 5.1. Measurement instrument and discriminant validity test

The loading factor (LF) of each instrument (item) was measured to assess the convergent validity of each item. To measure the validity, consistency, reliability, and multicollinearity defect, average variance extracted (AVE), Cronbach's alpha ( $\alpha$ ), composite reliability (CR), and variance inflation factor (VIF) of each construct was measured. The values of all these parameters were found within allowable range. The estimate of measurement instruments is shown in Table 2.

On estimation, it was found that the square roots of all the AVEs of the constructs are more than the concerned correlation bi-factor coefficients which confirms the discriminant validity in consonance with Fornell and Larcker criterion (Fornell & Larcker, 1981). To complement the result, the Heterotrait-Monotrait (HTMT) correlational ratio test was conducted (Henseler, Ringle, & Sarstedt, 2014), wherein it has been observed that the corresponding values of the constructs are all <0.85, confirming discriminant validity (Voorhees, Brady, Calantone, & Ramirez, 2016). The results are shown in Table 3 and in Table 4 respectively.

However, by calculating the cross-loading factors, discriminant validity may be explained by showing that all the cross-loading factors are less than the loading factors. The relevant results are shown in Appendix A2.

### 5.2. Common method variance (CMV)

CMV is affected by different aspects. These are procedures of survey measurement, context of survey, and choice of respondents (Rindfleisch, Malter, Ganesan, & Morman, 2008). In this context, it has been observed "some sources of common method biases result from the same source or rather, whereas others are produced by the measurement items themselves, the context of the items within the measurement instrument, and/or the context in which the measures are obtained" (Podsakoff, MacKenzie, Lee, & Podsakoff, 2003, p.881). Choice of survey respondents and the instrument design are under the control of the researchers. As such, it may have considerable impact on CMV. CMV may

**Table 2**  
Measurement instruments.

| Construct/ Instrument | LF   | AVE  | A    | CR   | VIF | t-value |
|-----------------------|------|------|------|------|-----|---------|
| ADM                   |      | 0.82 | 0.87 | 0.85 | 4.6 |         |
| ADM1                  | 0.88 |      |      |      |     | 21.16   |
| ADM2                  | 0.91 |      |      |      |     | 19.16   |
| ADM3                  | 0.97 |      |      |      |     | 31.78   |
| ADM4                  | 0.94 |      |      |      |     | 32.47   |
| ADM5                  | 0.84 |      |      |      |     | 31.41   |
| ADM6                  | 0.88 |      |      |      |     | 29.77   |
| OPE                   |      | 0.79 | 0.85 | 0.83 | 3.9 |         |
| OPE1                  | 0.84 |      |      |      |     | 19.91   |
| OPE2                  | 0.90 |      |      |      |     | 28.72   |
| OPE3                  | 0.96 |      |      |      |     | 26.41   |
| OPE4                  | 0.86 |      |      |      |     | 31.77   |
| OPE5                  | 0.88 |      |      |      |     | 29.06   |
| ACRM                  |      | 0.78 | 0.84 | 0.82 | 4.4 |         |
| ACRM1                 | 0.83 |      |      |      |     | 21.41   |
| ACRM2                 | 0.87 |      |      |      |     | 28.77   |
| ACRM3                 | 0.90 |      |      |      |     | 29.27   |
| ACRM4                 | 0.85 |      |      |      |     | 30.48   |
| ACRM5                 | 0.94 |      |      |      |     | 31.11   |
| ACRM6                 | 0.91 |      |      |      |     | 19.87   |
| ACRM7                 | 0.86 |      |      |      |     | 26.26   |
| ACRM8                 | 0.89 |      |      |      |     | 27.02   |
| BRS                   |      | 0.79 | 0.85 | 0.83 | 4.7 |         |
| BRS1                  | 0.84 |      |      |      |     | 19.47   |
| BRS2                  | 0.83 |      |      |      |     | 22.61   |
| BRS3                  | 0.88 |      |      |      |     | 31.71   |
| BRS4                  | 0.90 |      |      |      |     | 29.72   |
| BRS5                  | 0.96 |      |      |      |     | 26.41   |
| BRS6                  | 0.93 |      |      |      |     | 33.72   |
| FP                    |      | 0.81 | 0.86 | 0.84 | 3.8 |         |
| FP1                   | 0.88 |      |      |      |     | 26.88   |
| FP2                   | 0.87 |      |      |      |     | 28.11   |
| FP3                   | 0.83 |      |      |      |     | 29.21   |
| FP4                   | 0.91 |      |      |      |     | 19.47   |
| FP5                   | 0.97 |      |      |      |     | 26.12   |
| FP6                   | 0.95 |      |      |      |     | 27.11   |
| FP7                   | 0.90 |      |      |      |     | 30.07   |

**Note:** The details of the instruments are provided in Appendix A1.

be avoided had it been possible to use different respondents for different constructs. In the context of a questionnaire, it is good to use different response scales against different items. It is also better to arrange items at random. If we could take these steps, we could limit the potential of CMV. But our study is cross-sectional in nature. Each questionnaire was completed by a single respondent at a specific time. As such, our research is prone to CMV (Jap & Anderson, 2004). Therefore, to eliminate the chances of bias, we initially took some preemptive measures. We assured the potential respondents that their anonymity and confidentiality will be strictly preserved so that they can respond in an unbiased way. CMV could have been avoided to some extent by collecting responses from different respondents to eliminate person-specific bias. It is better to use two separate data sources. In a cross-sectional survey for measuring predictor variables and dependent variables, there is a need to use secondary data sources. This could limit the chance of CMV (Summers, 2001; Zinkhan, 2006; Hulland, Baumgartner, & Smith, 2018). These are the techniques to reduce CMV through the survey design. But such remedies are not always practical and feasible. Even when they are applied, their outcomes do not always seem to be sufficient. In those cases, we take recourse to statistical process (Simmering et al., 2015). Several methods can be used to diagnose CMV. These are (a) Harman's single factor test (SFT); (b) correlational procedure (partial) to control CMV in the scale-level (marker process); (c) Latent method factors for accounting CMV in the item-level; and (d) multiple method factors for correlating CMV. In our study, a post-hoc SFT, suggested by Harman, was conducted, where it was found that the first factor that emerged was 39.88%, which is within the threshold limit of 0.50 (Podsakoff et al., 2003). A marker variable technique (Lindell & Whitney, 2001) was followed, which highlighted that the differences in

**Table 3**

Discriminant validity test (Fornell and Larcker criterion).

| Construct | ADM    | OPE     | ACRM  | BRS    | FP   | AVE  | SD  | Mean |
|-----------|--------|---------|-------|--------|------|------|-----|------|
| ADM       | 0.91   |         |       |        |      | 0.82 | 1.4 | 3.4  |
| OPE       | 0.17   | 0.89    |       |        |      | 0.79 | 1.7 | 2.9  |
| ACRM      | 0.21** | 0.23    | 0.88  |        |      | 0.78 | 1.3 | 3.7  |
| BRS       | 0.18*  | 0.22    | 0.24* | 0.89   |      | 0.79 | 1.9 | 3.1  |
| FP        | 0.17   | 0.18*** | 0.26  | 0.23** | 0.90 | 0.81 | 1.7 | 2.7  |

**Table 4**

Discriminant validity test (HTMT criteria).

| Construct | ADM  | OPE  | ACRM | BRS  | FP |
|-----------|------|------|------|------|----|
| ADM       |      |      |      |      |    |
| OPE       | 0.37 |      |      |      |    |
| ACRM      | 0.41 | 0.32 |      |      |    |
| BRS       | 0.51 | 0.53 | 0.42 |      |    |
| FP        | 0.39 | 0.47 | 0.37 | 0.31 |    |

CMB and adjusted CMB for all the concerned constructs were <0.06, confirming the SFT.

Both these results indicate that there is no CMV in the data.

### 5.3. Multigroup analysis (MGA)

To verify the effects of the two moderators, TT and LS, on the linkages H1 and H2 (for TT) and on the linkage H4 (for LS), MGA was undertaken with the help of an accelerated and bias corrected bootstrapping approach with consideration of 5000 resamples. The moderator TT has been divided into two categories, which are High TT and Low TT, whereas the other moderator LS has been divided into two categories, which are Strong LS and Weak LS. For conducting MGA, a three-step process was undertaken (Matthews, 2017). In the first step, two categorical variables were selected. The variables are the groups which are large enough for statistical power. In the second step, the process of measurement invariance of composite models (MICOM) was followed (Henseler, Ringle, & Sarstedt, 2016) to determine if the measurement models have the same attributes under different conditions (Henseler, Ringle, & Sarstedt, 2015). This step is critical, and it is required to be addressed in the MGA process. The MICOM process helps to compare the group parameters and to identify whether there is any measurement invariance, partial measurement invariance, or any full measurement invariance (Henseler et al., 2016). In this study, for two categorical variables, the mean original differences fell between boundaries of 2.5% and 97.5%. Also, the original variance difference fell between the 2.5% to 97.5% range. Thus, full invariance is observed. Here, it should be mentioned that if the values of the mean or original variance difference come between the range of 2.5% and 97.5%, then according to the MICOM approach, it cannot further proceed with the MGA. In the final step, the results of statistical tests to compare the multigroup have been assessed. The SEM technique has several approaches that can be used for comparing the path coefficients (p-values) of the group. In this case, Hair, Sarstedt, Ringle, and Gudergan's (2017) permutation approach has been followed. The PLS algorithm and bootstrapping (5000 resamples) for each group, that is, for TT and for LS, has been computed separately. If the p-value output difference is < 0.10, it means that there is a significant difference between these groups. Thus, the p-value difference for the two categories of the moderator TT on ADM → BRS is 0.02; on OPE → BRS, it is 0.03; and for the moderator LS on BRS → FP, it is 0.02. All the differences are <0.10, which makes them significant. Hence, while conducting the MGA analysis, the step-by-step procedure was followed including the MICOM process. The entire results of multigroup analysis are encapsulated in Table 5.

**Table 5**

Multigroup analysis (MGA).

| Paths            | Moderators                 | p-value differences | Remarks     |
|------------------|----------------------------|---------------------|-------------|
| (ADM → BRS) × TT | Technology Turbulence (TT) | 0.02                | Significant |
| (OPE → BRS) × TT | Technology Turbulence (TT) | 0.03                | Significant |
| (BRS → FP) × LS  | Leadership Support (LS)    | 0.02                | Significant |

### 5.4. Assessment of structural equation modeling (SEM)

The proposed model has been examined using the blindfolding process along with the accelerated and bias-corrected bootstrapping approach with 6000 resamples. For this, omission separation of five was chosen to obtain cross-validated redundancy estimates for the relevant predicting factors (Lew, Sinkovics, Yamin, & Khan, 2016). Regarding Stone-Geisser  $Q^2$ , the value was found to be 0.68 (Geisser, 1975; Stone, 1974). It reveals that the results have predictive relevance. Again, to assess the model fit, Henseler et al.'s (2014) suggestion was taken. The standardized root mean square residual (SRMR) was considered as the index. The results show that the values of SRMR are 0.062 and 0.034 with PLS and with PLS<sub>c</sub> respectively, which are both less than the threshold highest value of 0.08 (Hu & Bentler, 1998), confirming that the model is in order. By following this process, path coefficients, p-values, and  $R^2$  values have been estimated. The validated model is shown in Fig. 3.

The estimates of path coefficients, p-values, and  $R^2$ -values are shown in Table 6.

### 5.5. Results

The study formulated seven hypotheses, out of which three (H5, H6, and H7) cover the effects of the moderators TT and LS on the linkages H1, H2, and H4. All these hypotheses have been found to be supported after validation. The ADM and OPE impact significantly the BRS, as the concerned path coefficients are 0.32, with a level of significance  $p < 0.01$  (\*\*) (H1), and 0.41, with level of significance  $p < 0.001$  (\*\*\*) (H2). The results highlight that between these two impacts, the influence of OPE on BRS (H2) is more, as the concerned path coefficient is greater (0.41). The ACRM significantly and positively impacts FP (H3), with the path coefficient being 0.53, having a level of significance of  $p < 0.001$  (\*\*\*). Moreover, BRS impacts FP (H4) positively and significantly, as the concerned path coefficient is 0.27, with a level of significance  $p < 0.001$  (\*\*\*). Concerning the effects of the two moderators, TT and LS, the results show that the effects of TT on the linkages covered by H1 and H2 are both negative, since the concerned path coefficients are −0.23, with a level of significance of  $p < 0.05$  (\*) on the linkage H1, and −0.32, having a level of significance of  $p < 0.001$  (\*\*\*). However, the effects of the other moderator LS on the linkage H4 are positive and significant, as the concerned path coefficient is 0.29, with a significance level of  $p < 0.05$  (\*).

So far as coefficients of determinants ( $R^2$ ) are concerned, from the results it can be inferred that ADM and OPE can explain BRS to the extent of 47%, whereas ACRM and BRS can explain FP to the tune of

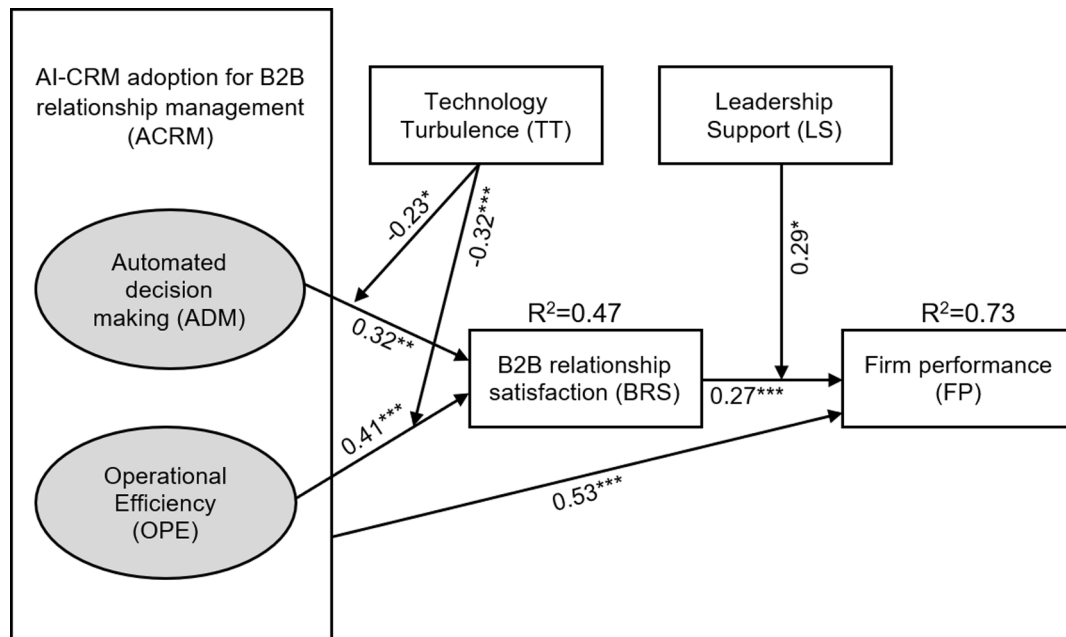


Fig. 3. Validated model.

**Table 6**  
Path coefficients, *p*-values along with *R*<sup>2</sup> values.

| Paths          | Hypotheses | Path coefficients / <i>R</i> <sup>2</sup> -values | <i>p</i> -values | Remarks   |
|----------------|------------|---------------------------------------------------|------------------|-----------|
| Effects on BRS |            | <i>R</i> <sup>2</sup> = 0.47                      |                  |           |
| By ADM         | H1         | 0.32                                              | **(<0.01)        | Supported |
| By OPE         | H2         | 0.41                                              | ***              | Supported |
| Effects on FP  |            | <i>R</i> <sup>2</sup> = 0.73                      |                  |           |
| By ACRM        | H3         | 0.53                                              | ***              | Supported |
| By BRS         | H4         | 0.27                                              | ***              | Supported |
| Effects of TT  |            |                                                   |                  |           |
| On H1          | H5         | −0.23                                             | *(<0.05)         | Supported |
| On H2          | H6         | −0.32                                             | ***              | Supported |
| Effects of LS  |            |                                                   |                  |           |
| On H4          | H7         | 0.29                                              | *(<0.05)         | Supported |

73%, which is the overall predictive power of the model. Incidentally, MGA also highlights that the effects of two moderators on the linkages concerned are significant.

### 5.6. Discussions on the results

This study has attempted to interpret and analyze how firm performance could be impacted by adopting an AI-integrated CRM system for B2B relationship management. By integrating AI with a CRM tool, it is possible to make more accurate decisions, because it automates analyzing the collected customer data. This impacts on the various firms involved in improving B2B relationships. This is the substance of H1, and it is supported by earlier studies (Alshare & Lane, 2011; Chatterjee et al., 2020a). By using an AI-CRM system in the B2B context, a firm's overall efficiency can be enhanced, affecting the satisfaction level of employees who managed B2B relationships, which answers hypothesis H2. Other studies also supported this idea (Lano, 2015; Molinillo & Japutra, 2017). Besides, applications of an AI-CRM system in the B2B

context improve the overall firm performance, as is evident from the validated hypothesis H3. This has been found in other studies (Inyang et al., 2018; Pandey & Mookerjee, 2018). This concept is found to have been supported by a study of Ko, Kim, Kim, and Woo (2008) that was conducted in the Korean fashion industry and analyzed how the characteristics of an organization impact its adoption of CRM. Again, if the firms involved in B2B relationships develop their activities to their satisfaction level, the performance of the firms will be improved, which validates H4. Other studies have also validated this concept (Keramati et al., 2010; Li et al., 2006). Moreover, whenever a firm introduces a new system, the employees usually resist its implementation, which is the main concept of the technology turbulence. It affects the process of developing B2B relationships among the firms. That is why technology turbulence impedes the satisfaction levels of employees involved in B2B relationships. This idea has also been supported by SQB theory (Samuelson & Zeckhauser, 1988). But if top management strongly supports adopting new technology like an AI-CRM system for a B2B context, the firm performance will not be hampered. That is why leadership support is found to have positively impacted the linkage H4.

Many studies have analyzed how an AI-CRM system, which a firm adopts, can impact firm performance (Chatterjee, 2019; Chatterjee et al., 2019, 2020a). This study has extended this idea to consider the applications of AI-CRM systems in the B2B context. Also, it is perceived that technology turbulence can impede employees' acceptance of new technology when it is introduced. It can be overcome if the firm's leadership actively and effectively supports the introduction.

In this study the effects of the two moderators TT and LS could be discussed through graphical representations. The effects of High TT and Low TT on the linkages H1 and H2 are shown in two graphs in Fig. 4.

In both graphs, continuous and dotted lines represent effects of High TT and Low TT respectively. The graphs show that, with the increase of ADM (for H1) and with the increase of OPE (for H2), BRS decreases as it is affected by High TT. Both graphs indicate that BRS increases in Low TT, as the gradients of the dotted lines are positive, but the gradients of the continuous lines are negative.

The effects of Strong LS and Weak LS on the linkage H4 are shown in a graph (Fig. 5).

Fig. 5 shows the effects of Strong LS and Weak LS through continuous and dotted lines respectively. The figure shows that, as BRS increases, the increase of FP is more with the effects of Strong LS than compared to

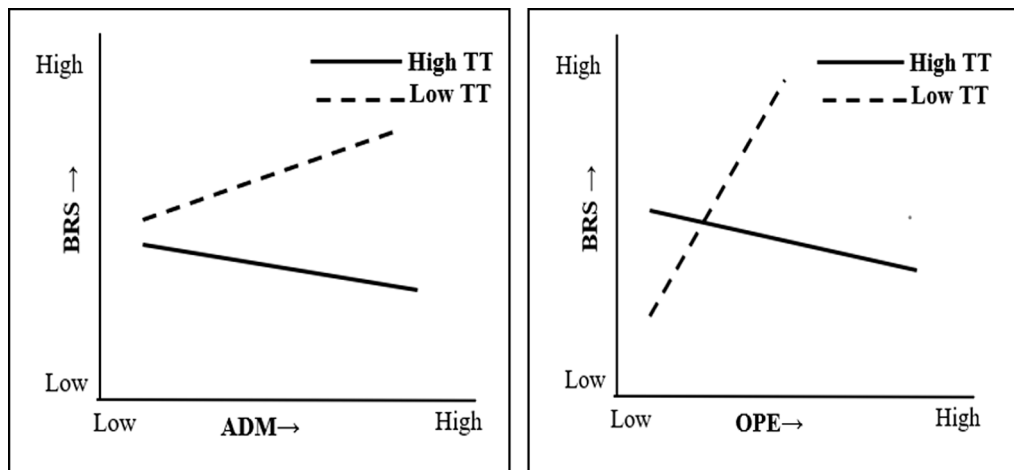


Fig. 4. Effects of TT on H1 and H2.

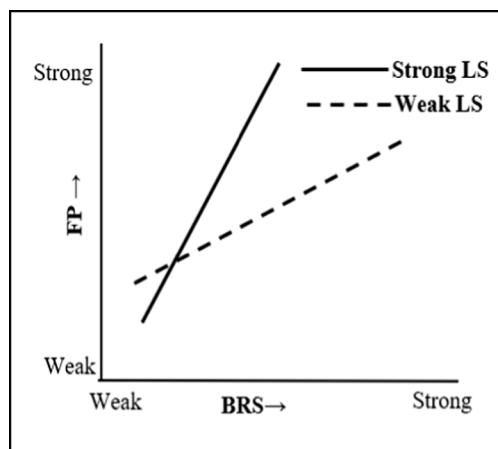


Fig. 5. Effects of LS on H4.

that of Weak LS, since the gradient of the continuous line is more than that of the dotted line.

## 6. Implications of the study

### 6.1. Theoretical contribution

This study makes several theoretical contributions to the extant literature. Several studies have analyzed how an AI-integrated CRM system affects firm performance (Chatterjee et al., 2019a; Chatterjee et al., 2020b). However, this is the first to have analyzed the effects an AI-integrated CRM system has on firm performance in the B2B context. This is also the first study to consider the effects of both technology turbulence and leadership support, as moderators, on firm performance when using an AI-CRM system in the B2B context. Many studies have used RBV theory (Kor & Mahoney, 2003), DCV (Teece et al., 1997), and SQB theory (Samuelson, & Zeckhauser, 1988), but ours is the first to use these theories to examine firm performance when applying an AI-integrated CRM system for B2B relationship management.

The present study claims to have provided empirical support for the notion that neither RBV theory nor DCV theory alone can address all the aspects to interpret a firm's capability, which is why the present study had to import the idea of SQB theory. By combining RBV, DCV, and SQB, we claim that the present study has been able to successfully interpret the different aspects of AI-CRM in the B2B relational scenario under the moderating effects of leadership support and technology turbulence. We

believe this is a special theoretical contribution of this study. Moreover, a study by Ullah et al. (2020) investigated how a firm adopting CRM could impact its performance with consideration of the moderating role of technology turbulence. The concept of Ullah et al.'s (2020) study has been extended in this present study to investigate how both technology turbulence and leadership support, as two moderators, influence an AI-integrated CRM system's impact on firm performance in the B2B context. This is claimed to have added value to extant literature. A study of Chang et al. (2010) demonstrated that, in the context of Korean companies, the application of CRM technology brings positive outcomes by strengthening marketing ability. The concept of that study has been extended in the present study to document how CRM tools embedded with AI technology could yield better firm performance. This has added value to the extant literature. Moreover, we could have chosen a standard adoption model to analyze how adopting an AI-CRM system for B2B relationship management impacts firm performance. Instead, we chose some better-suited factors in developing the theoretical model, which we claim to be successful, because the proposed theoretical model achieved a high predictive power of 73%. This is also claimed to be a unique theoretical contribution of this study.

### 6.2. Practical implications

Applications of AI for improving B2B relationships and, therefore, firm performance is gradually gaining momentum. Integration of AI with the existing CRM system or with a new CRM system for effectively managing B2B relationship is considered to be in great demand. The findings of this study provide useful guidelines for the leaders of firms regarding some salient points to consider before investing in AI for digitalization. The leaders should evaluate whether their organizations have the proper abilities to sense the dynamic changes which could help them also sense the external opportunities. They need to weigh if their organizations can seize the sensed opportunities and, finally, to assess if their organizations could integrate the seized external opportunities with the existing internal opportunities to ensure that their organizations are successfully AI-enabled. The study highlights that an AI-CRM system can help to improve B2B relationship management for quick decision making. Sometimes, a firm's B2B relationship manager might wonder which is the best way to contact the B2B counterpart of another firm. This decision can be automated through the CRM system integrated with AI. With appropriate applications of an AI-integrated CRM system, firms can automate most of the analysis and recommendations. Consequently, it will help the managers to make an appropriate decision at the right time, more accurately and cost-effective. That would eventually improve the operational efficiency of the firms. An AI-CRM system for B2B relationships will improve collaboration and coordination in

managing B2B relationships, thus enhancing satisfaction of the managers of the firms. Since the AI-integrated CRM system in the B2B relationship context impacts firm performance, as is seen from this study, top management needs to allocate adequate funds to support integrating AI with the CRM system and try to establish a good support system. Firms' top management and policymakers must articulate appropriate and implementable plans to train employees who will use the AI-CRM system, so that they can use the new system with confidence, thus overcoming the effects of technology turbulence. Adequately training the employees to use the AI-CRM system will improve B2B relationships. To support their employees' training, the leadership should provide training materials such as quick reference guides or freely available video tutorials.

Apart from all the above-mentioned practical contributions, the present study claims to have provided effective guidelines to a firm's leadership that considers investing in AI-CRM adoption for B2B use. The leadership should carefully evaluate its prevailing system's effects on its relationship abilities to assess if the firm is able to sense the dynamic changes in the internal and external environments, which is necessary to maintain the relationship with other firms and to improve the B2B relationship. The leadership is also required to evaluate if the firm's existing capabilities can seize the available opportunities by establishing cordial relationships with other firms. The leadership team must improve these capabilities before the firm adopts an AI-CRM for B2B relationship management, which will eventually help the firm to mitigate any constraint.

## 7. Conclusion, limitations of the study with direction for future scope

This study has focused on how adoption of an AI-integrated CRM system in the B2B context can influence firm performance. This study has developed a model based on this concept. While developing the model, we emphasized the importance of the B2B relationship in the context of using an AI-integrated CRM system amongst the firms involved. The model has achieved success, as its predictive power is as high as 73%. We have focused on improving the B2B relationship management in the dynamic environment by using AI-integrated CRM systems to enhance firm performance. The model shows that AI-CRM adoption for B2B relationship management, along with its other two components, ADM and OPE, has considerable impact on firm performance. Adopting AI-CRM for B2B relationship management alone can hardly improve the firm performance to the desired extent compared to that derived from additional consideration of automated decision making and operational efficiency. However, whatever may be the value of the proposed model, it cannot be said that it is free from all

limitations.

The present study is based on the inputs of the respondents of Indian firms. Hence, the results so derived are construed to be India-specific. The findings may not be equally applicable to other countries. Hence, the results lack generalizability. Also, the survey in the present study was conducted using inputs of 312 usable respondents, which seems to be an inadequate number to represent the entire society. The findings of the present study on how AI-CRM could impact B2B relationships were based on the analysis of data from respondents representing manufacturing and service firms only. This study did not consider representation of other types of firms, and thus, it is also a defect in its generalizability. The predictive power of the model is 73%. In this context, the present study appears to be silent on what steps have been taken to improve its explanative power. The present study is based on the findings from analyzing cross-sectional data, which creates problems in establishing accurate causality between the constructs. These are deemed to be salient limitations of this study.

To overcome these defects and to extend this study, we suggest that future researchers take some steps to rectify the limitations. The inputs of respondents spread across the globe could be considered to portray generalized results. Also, a survey could consider inputs from more respondents to represent the entire society. To project generalizable results, a future study should include representation from agricultural firms and conglomerates as well as from service and manufacturing firms. To enhance the explanative power of the model, other relevant boundary conditions could be examined to assess if they could strengthen the model. To remove the defects arising from utilizing cross-sectional data, a longitudinal study could be undertaken with econometric analysis, which could remove the causality and endogeneity defects.

## CRediT authorship contribution statement

**Sheshadri Chatterjee:** Resources, Investigation. **Ranjan Chaudhuri:** Data curation. **Demetris Vrontis:** Conceptualization.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgement

None.

## Appendix

### Appendix A1. Measurement items with sources

| Items | Source(s)                                        | Statements                                                                                                 | Response<br>[SD][D][N][A]<br>[SA] |
|-------|--------------------------------------------------|------------------------------------------------------------------------------------------------------------|-----------------------------------|
| ADM1  | Verma & Verma, 2013                              | AI-CRM helps us with decision-making process.                                                              | [1][2][3][4][5]                   |
| ADM2  | Chatterjee et al., 2020b                         | Appropriate algorithms need to be applied with AI-CRM technology for correct auto decision-making process. | [1][2][3][4][5]                   |
| ADM3  | Irsic, 2017                                      | Accurate decisions can be made using the AI-CRM system.                                                    | [1][2][3][4][5]                   |
| ADM4  | Alshare & Lane, 2011                             | AI-CRM helps better decision making for managing B2B relationships.                                        | [1][2][3][4][5]                   |
| ADM5  | Chatterjee et al., 2019a; Davenport et al., 2020 | AI-CRM supplements manual decision-making process for maintaining B2B relationships.                       | [1][2][3][4][5]                   |
| ADM6  | Chatterjee et al., 2020a                         | I would recommend other firms to use AI-CRM for better decision making.                                    | [1][2][3][4][5]                   |
| OPE1  | Nguyen & Mutum, 2012                             | AI integrated with traditional CRM system helps managing huge volume of data                               | [1][2][3][4][5]                   |
| OPE2  | Lano, 2015; Agnihotri et al., 2016               | I believe that AI integrated CRM system improves operational efficiency of the firms.                      | [1][2][3][4][5]                   |
| OPE3  | Graca, Barry, & Doney, 2015                      | Our firm uses the AI-CRM system for contact management process.                                            | [1][2][3][4][5]                   |

(continued on next page)

(continued)

| Items | Source(s)                                                       | Statements                                                                                                                            | Response<br>[SD][D][N][A]<br>[SA] |
|-------|-----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| OPE4  | Li & Nguyen, 2016                                               | AI-CRM helps in auto-recommendation for appropriate actions to be taken.                                                              | [1][2][3][4][5]                   |
| OPE5  | Molinillo & Japutra, 2017                                       | I think that AI-CRM helps in cost saving to the firms.                                                                                | [1][2][3][4][5]                   |
| ACRM1 | Mustak, 2014                                                    | Adoption of AI-CRM system makes our firm more agile.                                                                                  | [1][2][3][4][5]                   |
| ACRM2 | Lipiaainen, 2015                                                | Not all the departments in our firm use the AI-CRM system.                                                                            | [1][2][3][4][5]                   |
| ACRM3 | Ascarza et al., 2018                                            | Our firm uses hybrid model by using AI-CRM and Traditional CRM system.                                                                | [1][2][3][4][5]                   |
| ACRM4 | Chatterjee et al., 2019a                                        | I think employee-training is important to improve adoption of AI-CRM in the firm.                                                     | [1][2][3][4][5]                   |
| ACRM5 | De Luca et al., 2020                                            | I would recommend using AI-CRM than traditional CRM for managing B2B relationships.                                                   | [1][2][3][4][5]                   |
| ACRM6 | Alshawi et al., 2011; Chatterjee et al., 2020a                  | Employees would like to use AI-CRM system as they get auto-recommendations at the right time.                                         | [1][2][3][4][5]                   |
| ACRM7 | Davenport et al., 2020                                          | Further improvement of features and functionalities of AI-CRM system is necessary for increasing the adoption of AI-CRM system.       | [1][2][3][4][5]                   |
| ACRM8 | Agnihotri et al., 2016                                          | I believe that adoption of AI-CRM solution for B2B relationship management is still in the initial stage in our firm.                 | [1][2][3][4][5]                   |
| BRS1  | Dans Perez, 2001                                                | I think employees involved in B2B relationship will be more satisfied using AI integrated CRM system than traditional CRM tool.       | [1][2][3][4][5]                   |
| BRS2  | Firth & Swanson, 2001                                           | I believe that AI-CRM system can help better manage the B2B relationship.                                                             | [1][2][3][4][5]                   |
| BRS3  | Sin et al., 2005; Saura, Palos-Sanchez, & Blanco-González, 2019 | I think that AI-CRM can help in addressing customer queries more accurately.                                                          | [1][2][3][4][5]                   |
| BRS4  | Li et al., 2006; Chatterjee et al., 2019a                       | I trust that AI-CRM system helps improve B2B relationship satisfaction.                                                               | [1][2][3][4][5]                   |
| BRS5  | Keramati, Mehrabi, & Mojir, 2010                                | Appropriate usage of AI-CRM system is necessary for managing B2B relationship along with the support of leadership team of the firms. | [1][2][3][4][5]                   |
| BRS6  | Ahearne et al., 2012                                            | AI-CRM can help improve B2B relationships by increasing customer loyalty.                                                             | [1][2][3][4][5]                   |
| FP1   | Rosomme, 2003; Li et al., 2006                                  | I believe that our sales growth has improved post AI-CRM implementation.                                                              | [1][2][3][4][5]                   |
| FP2   | Keramati, Mehrabi, & Mojir, 2010                                | Using AI-CRM system can help to get all the relevant reports on time.                                                                 | [1][2][3][4][5]                   |
| FP3   | Dans Perez, 2001; Ahearne et al., 2012                          | Using AI-CRM system for B2B relationship management can help improve competitiveness of the firms.                                    | [1][2][3][4][5]                   |
| FP4   | Firth & Swanson, 2001                                           | I believe that AI-CRM system helps in quick decision-making process which improves firm performance.                                  | [1][2][3][4][5]                   |
| FP5   | Sin et al., 2005                                                | I think we have received more orders from our partners post AI-CRM deployment in our firm.                                            | [1][2][3][4][5]                   |
| FP6   | Chatterjee et al., 2019a                                        | I believe AI-CRM can improve B2B relationship which can help improve our firm image.                                                  | [1][2][3][4][5]                   |
| FP7   | Saura, Palos-Sanchez, & Blanco-González, 2019                   | I believe that post AI-CRM implementation, our firm will increase its market share.                                                   | [1][2][3][4][5]                   |

SD = Strongly Disagree; D = Disagree; N = Neither agree nor disagree; A = Agree; SA = Strongly Agree.

*Appendix A2. Loading and cross-loading estimates for discriminant validity test*

Loadings and cross loadings have been estimated and are shown in the following table. The results show that all the cross loadings are less than the values of loadings. It confirms discriminant validity.

| Constructs/ Items | ADM  | OPE  | ACRM | BRS  | FP   |
|-------------------|------|------|------|------|------|
| ADM1              | 0.88 | 0.14 | 0.17 | 0.19 | 0.21 |
| ADM2              | 0.91 | 0.16 | 0.31 | 0.21 | 0.26 |
| ADM3              | 0.97 | 0.22 | 0.32 | 0.32 | 0.31 |
| ADM4              | 0.94 | 0.26 | 0.25 | 0.34 | 0.30 |
| ADM5              | 0.84 | 0.21 | 0.28 | 0.36 | 0.22 |
| ADM6              | 0.88 | 0.34 | 0.29 | 0.39 | 0.20 |
| OPE1              | 0.13 | 0.84 | 0.17 | 0.37 | 0.24 |
| OPE2              | 0.17 | 0.90 | 0.16 | 0.36 | 0.26 |
| OPE3              | 0.15 | 0.96 | 0.24 | 0.15 | 0.28 |
| OPE4              | 0.29 | 0.86 | 0.22 | 0.27 | 0.23 |
| OPE5              | 0.19 | 0.88 | 0.31 | 0.24 | 0.31 |
| ACRM1             | 0.21 | 0.27 | 0.83 | 0.27 | 0.18 |
| ACRM2             | 0.26 | 0.22 | 0.87 | 0.32 | 0.26 |
| ACRM3             | 0.25 | 0.23 | 0.90 | 0.18 | 0.28 |
| ACRM4             | 0.31 | 0.26 | 0.85 | 0.19 | 0.33 |
| ACRM5             | 0.31 | 0.32 | 0.84 | 0.24 | 0.19 |
| ACRM6             | 0.16 | 0.34 | 0.91 | 0.17 | 0.27 |
| ACRM7             | 0.21 | 0.37 | 0.86 | 0.21 | 0.24 |
| ACRM8             | 0.26 | 0.19 | 0.89 | 0.26 | 0.31 |
| BRS1              | 0.17 | 0.31 | 0.32 | 0.84 | 0.14 |
| BRS2              | 0.22 | 0.33 | 0.17 | 0.83 | 0.17 |
| BRS3              | 0.20 | 0.30 | 0.28 | 0.88 | 0.26 |
| BRS4              | 0.31 | 0.21 | 0.15 | 0.90 | 0.22 |
| BRS5              | 0.21 | 0.18 | 0.17 | 0.96 | 0.21 |
| BRS6              | 0.26 | 0.17 | 0.33 | 0.93 | 0.24 |
| FP1               | 0.14 | 0.19 | 0.22 | 0.14 | 0.86 |
| FP2               | 0.17 | 0.32 | 0.27 | 0.17 | 0.87 |
| FP3               | 0.22 | 0.21 | 0.28 | 0.22 | 0.83 |
| FP4               | 0.19 | 0.26 | 0.17 | 0.24 | 0.91 |

(continued on next page)

(continued)

| Constructs/ Items | ADM  | OPE  | ACRM | BRS  | FP   |
|-------------------|------|------|------|------|------|
| FP5               | 0.28 | 0.34 | 0.26 | 0.27 | 0.97 |
| FP6               | 0.16 | 0.23 | 0.25 | 0.31 | 0.95 |
| FP7               | 0.31 | 0.17 | 0.19 | 0.33 | 0.90 |

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